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After completing the preparation materials, students should be able to:

1. **PK properties → onset and duration of action**

Relate the pharmacokinetic properties of the **ganglion blocking drugs** and **the neuromuscular blocking agents (NMBAs)** to their onset and duration of action in patients with full hepatic and renal function and those with impaired hepatic and renal function by giving specific examples.

2. **Structure-activity relationship → Site of action and Mechanism of action**

Explain how the structural features of the **ganglion blocking drugs** and the **NMBAs** determine relative specificity for neuronal or neuromuscular nicotinic receptors, respectively.

3. **Site of action / Mechanism of action → Therapeutic uses and Adverse effects**

– **Ganglion blockers:** Predict the therapeutic and adverse effects of ganglion nicotinic receptor blockade on organ systems exemplified by mecamylamine.

– **NMBAs:** Differentiate the actions and effects of depolarizing neuromuscular blockade by succinylcholine and the nondepolarizing NMBAs.

4. **NMBA reversal**

Explain the mechanisms of the cholinesterase inhibitors and sugamadex for reversing neuromuscular block and their potential adverse effects including specific examples.

5. **Mechanisms and effects → Drug-drug and drug-disease interactions**

Explain the mechanisms and effects of drug-drug and drug-disease interactions of the ganglion blockers and NMBAs.

Key points: PK properties → onset and duration of action

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Ganglion blocker

Mecamylamine	oral, complete absorption, crosses BBB, urine/unchanged drug, onset 30-120 min; duration 6->12 h; dosing 2-4x daily based on BP response

NMBAs: all IV; polar – do not cross BBB; onset and duration depend on depend on age and dose

Succinylcholine	IV/M; rapidly hydrolized by plasma cholinesterase to inactive metabs. rapid onset (IV <60 sec; IM 2-3 min), short duration (IV <8 min; IM 10-20 min) • Patients with deficient or absent plasma cholinesterase (rare) have increased duration of neuromuscular block, which could lead to prolonged apnea.
Benzylisoquinolines	
Atracurium	ester hydrolysis and Hofmann elimination, laudanosine metabolite onset in minutes, recovery begins in ~30-45 min
Cisatracurium	Hofmann elimination, onset in minutes, recovery begins in ~30-45 min
Aminosteroids	
Rocuronium	biliary/feces excretion, onset 1-2 min, recovery 20-120 min
Vecuronium	bilary and renal excretion, onset 2-4 min, recovery begins in ~20-40 min

Key points:

Structure-activity relationship → Site of action and Mechanism of Action

Ganglion blockers are synthetic amines with structural characteristics similar to acetylcholine. They are competitive inhibitors of neuronal nicotinic acetylcholine receptors (nAChRs). Ganglion blockers have lower affinity for neuromuscular nAChRs.

- Mecamylamine is the only one currently in clinical use. It's bulky bicyclic structure and secondary amine are necessary for interaction with neuronal nAChRs.
- Ganglion blockers block the actions of acetylcholine at nicotinic receptors – parasympathetic and sympathetic autonomic ganglia and the adrenal medulla – which blocks the entire output of the ANS. Mecamylamine is lipophilic, crosses the blood-brain barrier and also inhibits central nAChRs.

Neuromuscular blocking agents (NMBAs) also have structural characteristics similar to acetylcholine. They all have one or two quaternary ammoniums. NMBA's currently in clinical use have lower affinity for neuronal nAChRs.

- Succinylcholine is a **depolarizing** NMBA with a linear structure consisting of two acetylcholine molecules. Succinylcholine initially opens the channel, causing depolarization of the motor end plate and transient contractions. Succinylcholine is not metabolized in the synapse so the membranes remain depolarized and unresponsive to subsequent impulses = depolarizing blockade.
- The **nondepolarizing** NMBAs have a bulky, semi-rigid ring structure. They are subcategorized according their structures – isoquinolines and aminosteroids. These drugs are competitive inhibitors of acetycholine. In larger doses they can enter the channel pore, producing a more intense blockade.

Key points:
Mechanism of action / Site of action → Therapeutic uses and Adverse effects

Ganglion blockers:

Mecamylamine is used in the treatment of moderate to severe hypertension and uncomplicated malignant hypertension. Many drugs with more selective actions and fewer side effects are available.

The adverse effects are related to the physiologic responses. The physiologic responses are related to:

- 1) the responses of the effector organs to autonomic nerve impulse, and
- 2) the division of the autonomic nervous system exerting the dominant tone at the various organs.
- 3) the homeostatic reflexes that regulate autonomic tone (activity) are *absent*.

When drugs block the ganglion nAChRs, the prevailing (dominant) parasympathetic or sympathetic tone is reversed.

- Responses are complex and mostly unpredictable.
- All organs of the autonomic nervous system are affected. Please see the table on slide #16.
- Mecamylamine causes severe orthostatic hypotension (dizziness, lightheadedness, syncope), constipation, and urinary retention.

Key points:
Mechanism of action / Site of action → Therapeutic uses and Adverse effects

Neuromuscular nicotinic receptor blockers:

- NMBAs block transmission of the end plate potential, resulting in skeletal muscle paralysis (muscle relaxation), which facilitates endotracheal intubation and surgical procedures. They may be used to assist mechanical ventilation in the intubated patient.
- The respiratory muscles (intercostals and diaphragm) are the last to relax and the first to recover.
- Succinylcholine is useful for rapid endotracheal intubation and procedures requiring brief muscle relaxation, such as electroconvulsive therapy, because of its rapid onset of action and short duration of action.
- Adjunctive use of nondepolarizing NMBAs provides adequate muscle relaxation for all types of surgical procedures without the cardiorespiratory depressant effects produced by deep anesthesia.

  Prior to the introduction of NMBAs, only deep anesthesia with inhaled anesthetics that caused profound depressant effects on cardiovascular and respiratory systems would produce adequate muscle relaxation.
- Succinylcholine has numerous potential adverse effects. It can cause ventricular dysrhythmia, cardiac arrest, and death from hyperkalemia in pediatric patients who were subsequently found to have undiagnosed skeletal muscular dystrophy. The major adverse effect of nondepolarizing NMBAs is prolonged paralysis, respiratory depression, and apnea.
- Age: Neonates may be more sensitive to the effects. Older patients may have reduced clearance.
- Hypersensitivity reactions occur rarely.

NMBA reversal agents

Cholinesterase inhibitors: Neostigmine or Pyridostigmine

- Increase the concentration of ACh in the neuromuscular junction → ACh outcompetes the NMBA at the nAChR → prompt reversal of NMBA-induced skeletal muscle paralysis
- They reverse nondepolarizing neuromuscular blockade by the isoquinolines and aminosteroids.
- Large doses of cholinesterase inhibitor when neuromuscular block is minimal (near recovery) can result in neuromuscular dysfunction due to excessive acetylcholine in the neuromuscular junction.

Sugammadex – rapid reversal:

- Chemical chelator (γ-cyclodextrin) forms a complex in plasma with the aminosteroids, rocuronium and vecuronium (but not the benzylisoquinolines), in a 1:1 ratio. The complex is eliminated in the urine.

Adverse effects:

- May cause bradycardia and ECG abnormalities (monitor during drug administration)
- Serious hypersensitivity reactions / anaphylaxis reported (uncommon)
- Increases in coagulation parameters; clinically significant bleeding not reported
- Sugammadex may decrease plasma concentrations of hormonal contraceptive by steroid encapsulation.

Succinylcholine reversal by diffusion:

- Spontaneous reversal as SCh diffuses away from synapse
- Cholinesterase inhibitors augment depolarizing block by increasing ACh concentration in the synapse, which contributes to the depolarizing block.

Key points: Mechanisms and effects → Drug-drug and drug-disease interactions

Interaction with ganglion blocker	Effect
Drugs acting at autonomic receptors on effector organs	Enhance or reverse BP lowering effect
Drugs with the same adverse effects	Enhance CNS and peripheral side effects

Interaction with NMBAs	Effect
Volatile halogenated anesthetics	Enhance NM blockade Desirable – Adjunctive NMBA allows lower anesthesia concentration.
Aminoglycoside antibiotics Fluoroquinolone antibiotics Tetracyclines	Potentiate NM blockade: · Interference with ACh release in the NM junction and/or interfering with receptor function → prolonged muscle weakness and respiratory depression
Many other drugs may cause significant adverse events	Mechanisms: Potentiation or inhibition of effect Enhance respiratory depressant effects Enhance other adverse effects Contribute to electrolyte imbalance
Neuromuscular autoimmune disease Myasthenia gravis Lambert-Eaton myasthenic syndrome	Decreased NM nicotinic receptors (MG) Decreased ACh release (LEMS) potentiate NMBAs' effects.

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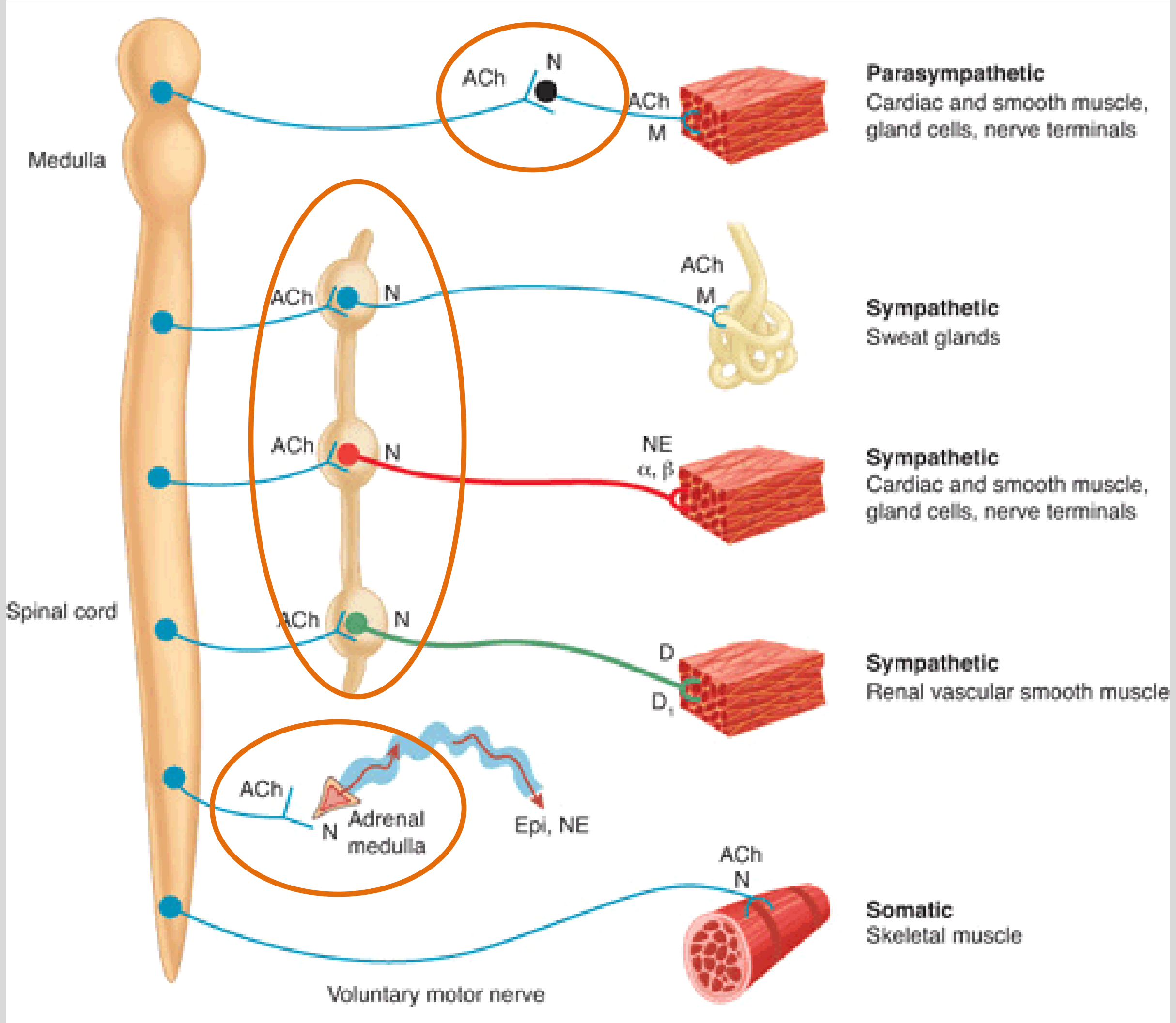


Ganglion Blockers

– drugs that block neuronal nicotinic acetylcholine receptors (nAChRs)

✳ Mecamylamine

- Trimethaphan (historical mention)
- Hexamethonium (research use)

Ganglion blockers are rarely used therapeutically due to myriad side effects.

➢ If ganglion blockers are not often used, why are they included in pharmacology lectures?

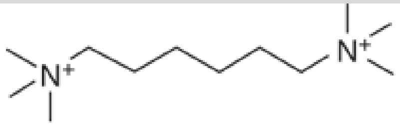
Understanding the effects of ganglion blockers requires a deep knowledge of the autonomic nervous system, including ANS regulation of heart rate, blood pressure, gastrointestinal motility, and ocular dynamics.

Grasping the concepts means students can appreciate how ganglion blockers influence various physiological processes and manage the potential side effects and clinical implications effectively.

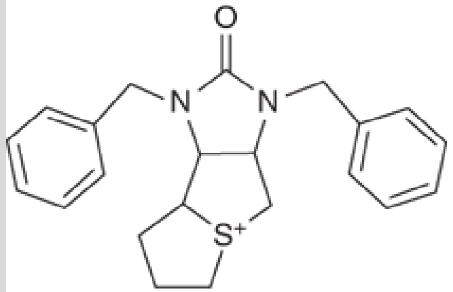
Ganglion Blocking Drugs:

NON-depolarizing neuronal nAChR block of receptors → block ganglionic transmission of **parasympathetic** and **sympathetic** autonomic ganglia and adrenal medulla.

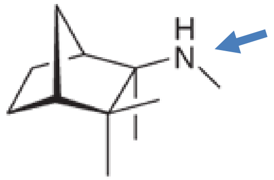
Compare to nicotine, which causes dose-dependent DEpolarizing block of all nicotinic receptors.



HEXAMETHONIUM (C6)



TRIMETHAPHAN



MECAMYLAMINE

Source: Laurence L. Brunton, Björn C. Knollmann: *Goodman & Gilman's: The Pharmacological Basis of The*

Mecamylamine: lipophilic
↓
distributes to CNS

Mecamylamine:

Mechanism: Channel plug – blocks the open channel pore of the nAChR complex.

The channel closes around the drug when the agonist unbinds trapping the drug molecule inside. Persistent block cannot be overcome by adding more agonist.

Structure-activity relationship: uncharged and lipophilic

- Rigid bicyclic structure fits the binding site of nAChRs.
- The secondary amine (arrow) is critical for interaction with the acetylcholine receptor by forming hydrogen bonds.
- Lower affinity for neuromuscular nAChRs.

Pharmacokinetics

- Oral, excreted unchanged in urine, duration 6->12h
- Mecamylamine crosses the blood-brain barrier and *inhibits central nAChRs*.
- Weak base: Acidic urine increases rate of excretion and may decrease therapeutic effect.

- Trimethaphan: competitive antagonist, does not cross BBB
- Hexamethonium: blocks the nicotinic ion channel after it is open

Responses to treatment with ganglion blockers are complex and largely unpredictable. (Details on the next slide.)

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Ganglion blockers…

- suppress outflow from nAChRs of both parasympathetic and sympathetic ganglia and adrenal medulla
- **suppress of all autonomic outflow**

Ganglion blockers…

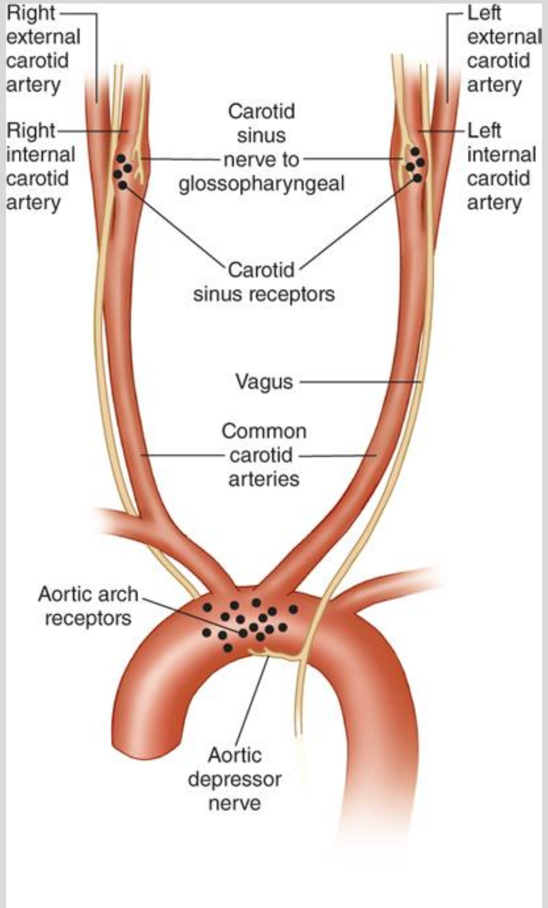
- suppress homeostatic reflexes that normally moderate autonomic responses
- **suppress normal cardiac, vascular, neurologic, muscle, and neurohumoral reflexes**

Mechanism of baroreceptor signaling (which is suppressed by ganglion blockers)**:**

- The ganglia are relay points in the ANS.
- They transmit signals from the baroreceptors to the brain and back to the effector organs.
- When they are disrupted, the signaling is prevented.

Examples:

- Rapid fall in blood pressure sensed by baroreceptors sends signals to the medulla oblongata → the sympathetic nervous system is disinhibited (is increased) → stimulates heart rate and vasoconstriction to increase blood pressure.
- Rapid rise in blood pressure sensed by baroreceptors sends signals to the medulla oblongata → parasympathetic nervous system activity increases → slows the heart rate and stimulates vasodilation to lower blood pressure.



Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan: Ganong's Review of Medical Physiology, Twenty-Sixth Edition

Mecamylamine Uses and Cautions

Therapeutic Uses

- Chronic hypertension
- Malignant hypertension (uncomplicated)

Lack of selectivity causes a broad range of undesirable effects.

- **Mecamylamine causes orthostatic hypotension** (absent baroreceptor reflexes) → reduced blood flow to brain → dizziness, lightheadedness, syncope

Drug interactions *High potential*

- Patients receiving ganglionic blockers are fully responsive to autonomic drugs acting on autonomic receptors at the effector organs: mAChRs and at αARs and βARs → exaggerated or reversed responses because the homeostatic reflexes are absent.
- Adverse effects may be enhanced by other drugs producing the same side effects.

Contraindications	Mechanism
Coronary insufficiency	Reduced SNS regulation of heart function and blood vessel tone
Recent myocardial infarction	
Pyloric stenosis	Gastric stasis, worsening nausea, vomiting
Glaucoma	Cycloplegia →↓ aqueous humor outflow →↑ intraocular pressure
Uremia	Reduced blood flow to kidney → kidney impairment → waste products accumulate
Unreliable, uncooperative patients	Mecamylamine dosing can be complicated, therapy must be carefully monitored, and the drugs cause orthostatic hypotension.

Check your knowledge.

1. **What is the overall effect of ganglion blockade by mecamylamine?**

2. **What are the mechanism of orthostatic hypotension and symptoms caused by mecamylamine?**

3. **Why should mecamylamine be prescribed only for patients who are reliable and cooperative with the therapy?**

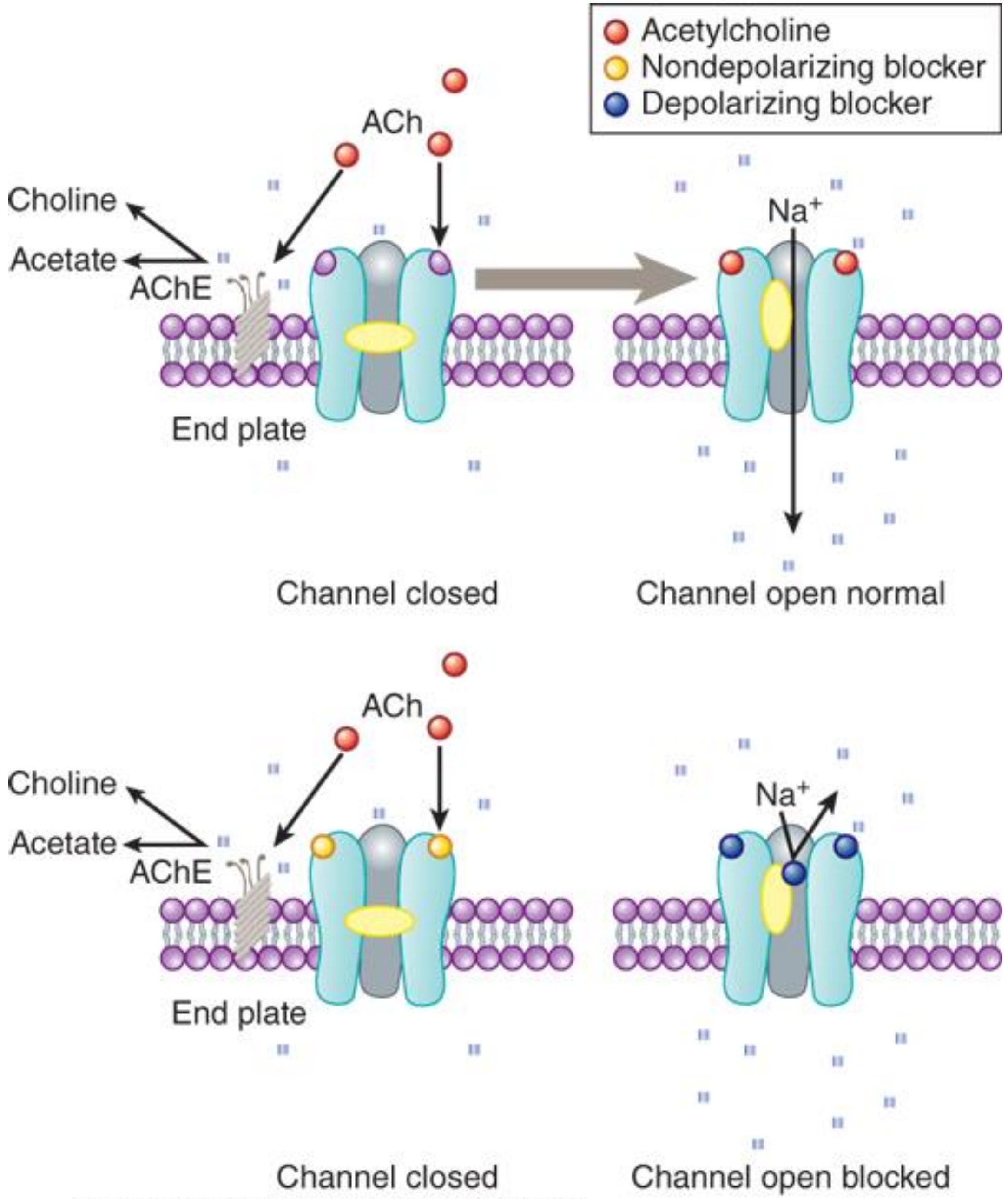
4. **Since mecamylamine is rarely used, why is it important for students to learn about it (other than because it might appear on a board exam?**

Pharmacokinetics

Drug	Elimination	Onset	Duration (95% recovery)	Potency*
BENZYLISOQUINOLINES Intermediate-acting:				
Atracurium	Ester hydrolysis (hepatic) and spontaneous Hofmann elimination (nonenzymatic, independent of liver/kidney function) · Laudanosine byproduct, CNS stimulant · Longer half-life than atracurium's half-life · May accumulate with long infusions (days)	2-3 min	20-35 min (90 min)	1.5
Cisatracurium cis-isomer of atracurium	Hofmann elimination	2-3 min	25-44 (~30-90)	1.5
AMINOSTEROIDS				
Intermediate-acting:				
Rocuronium	Biliary primarily, minor renal excretion	1-2 min	20-120 min	0.8
Vecuronium	Biliary (40-75%) and renal (~30%)	2.5-3 min	20-40 min (~45-65 min)	6
Pancuronium is no longer available in US or Canada. Long-acting; causes tachycardia due to sympathomimetic stimulation and cardiac muscarinic receptor blockade; high incidence of postop residual neuromuscular weakness				

*Potency is compared to tubocurarine potency = 1.

Plasma clearance, volume of distribution, and elimination half-lives may be influenced by patient age, volatile anesthetics, and the presence of renal or hepatic disease.



Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition

Schematic diagram of the interactions of drugs with the acetylcholine receptor on the end plate channel (structures are purely symbolic).

Top: The action of the normal agonist, acetylcholine in opening the channel (red spheres labeled ACh).

Bottom, left: A nondepolarizing blocker, eg, rocuronium, is shown as preventing the opening of the channel when it binds to the receptor (depicted as small yellow spheres occupying active sites).

Larger doses of the nondepolarizing NMBAs → the drugs enter the channel pores → more intense motor blockade (not shown)

Bottom, right: A depolarizing blocker, eg, succinylcholine, both occupying the receptor and blocking the channel. Normal closure of the channel gate is prevented, and the blocker may move rapidly in and out of the pore (shown as blue spheres occupying active sites and inside the pore).

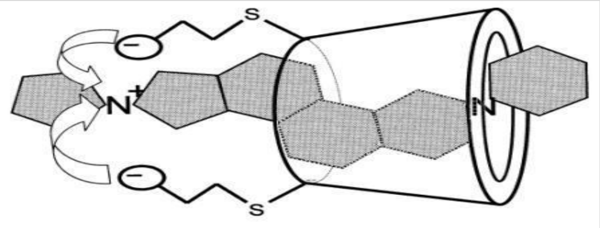
Depolarizing blockers may desensitize the end plate by occupying the receptor and causing persistent depolarization.

An additional effect of drugs on the end plate channel may occur through changes in the lipid environment surrounding the channel (not shown). General anesthetics and alcohols may impair neuromuscular transmission by this mechanism.

Slide adapted by LG

Reversal of nondepolarizing NMBA-induced muscle relaxation

Isoquinolines Atracurium, Cisatracurium	**Aminosteroids** Rocuronium, Vecuronium
Cholinesterase inhibitor – **neostigmine** or pyridostigmine – Augments ACh concentration in the synapse to outcompete the nondepolarizing blocker Neostigmine: faster onset of action than pyridostigmine, a shorter duration of action, and generally considered more effective.	**Sugammadex** – Chelating agent forms a complex with steroid NMBA in a 1:1 ratio → rapid ↓NMBA in levels – Complex is excreted in urine. – Cholinesterase inhibitors are also effective reversal agents.



Note:

Cholinesterase inhibitors are administered with atropine or glycopyrrolate to prevent vagal bradycardia.

Large doses of cholinesterase inhibitor administered when neuromuscular block is minimal (near recovery) can result in neuromuscular dysfunction due to excessive acetylcholine in the neuromuscular junction.

Sugammadex adverse effects: Serious adverse effects are rare.

Bradycardia, ECG abnormalities (monitor during therapy)

May encapsulate steroid contraceptives and reduce effectiveness of the contraceptive.

May increase coagulation parameters; clinically significant bleeding has not been reported.

NMBAs Class Adverse Effects and Cautions

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Prolonged apnea, risk factors Inadequate reversal, residual NM block, prolonged surgery, opioids
Electrolyte abnormalities may potentiate block Acidosis, hypokalemia, hypocalcemia, hypermagnesemia
Burns, Skeletal muscle trauma Increased doses of NMBA required
Neuromuscular diseases enhance block Myasthenia gravis: Autoantibodies directed against the NM nicotinic receptors Lambert-Easton myasthenic syndrome: Autoantibodies directed against voltage-gated Ca^{2+} channels, which decreases release of ACh.
Age Neonates may have increased sensitivity to block. Older patients may have reduced hepatic / renal function.
Hypersensitivity reactions are rare
Drug interactions: Aminoglycosides and other antibiotics increase ACh release or interfere with receptor function and prolong muscle weakness and respiratory depression Many other drugs: Potentiate or inhibit NMBA effects, enhance respiratory depression, or contribute to electrolyte imbalance.

Comparing the nondepolarizing NMBAs
(The takeaways are in red.)

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Atracurium	Cisatracurium	Rocuronium	Vecuronium
Hemodynamic effects: · **Histamine** release causes skin flushing, hypotension, tachycardia **Metabolism:** · plasma esterase hydrolysis and Hofmann elimination → **laudanosine** · high concentrations → accumulation (over days) → can cause **CNS excitation and lower seizure** · → Dosage adjustment of inhaled anesthetic may be required. No dose adjustment needed for liver or kidney impairment	Minimal hemodynamic effects: · Does not release histamine and not vaogolytic Metabolized by Hofmann elimination · more potent → lower dose compared to atracurium → low accumulation of laudanosine · **not likely to cause significant toxicity due to laudanosine** No dosage adjustment for kidney or liver impairment necessary	Minimal hemodynamic effects: · Does not release histamine and not vagolytic **Quick onset:** · **Least potent** of the nondepolarizing NMBAs in clinical use → rapid onset · **Often used for rapid sequence intubation** Excreted in bile, minor excretion in urine · Suitable for patients with renal impairment · Dose adjustment may be needed in patients with liver disease	Minimal hemodynamic effects: · Does not release histamine and not vagolytic >6 times more potent than rocuronium · Slightly slower onset than rocuronium · **not suitable for rapid sequence intubation** **Hepatic metabolism and excretion through both biliary and renal routes** · Patients with liver or kidney impairment may have prolonged response

Pancuronium: No longer available in US or Canada. Long-acting; tachycardia due to sympathomimetic stimulation and cardiac muscarinic receptor blockade; High incidence of postop residual neuromuscular weakness

Depolarizing NMBA
✳Succinylcholine

Drug	Elimination	Onset	Duration	Potency
Succinylcholine	Plasma cholinesterase Only a small fraction of the dose reaches the motor end plate.	0.5 min	<8 min	0.4 tubocurarine = 1
	Pharmacogenomics: Duration is prolonged in patients with deficient or absent plasma cholinesterase. The *BCHE* genetic variants are rare.			

- Succinylcholine has a very short duration because it is rapidly hydrolyzed by liver butyrylcholinesterase and plasma cholinesterase (mainly plasma cholinesterase). Only a small fraction of the dose reaches the neuromuscular junction.
- Succinylcholine is not degraded by AChE.
- **Succinylcholine blockade is terminated by diffusion** of the drug from the motor end plate into extracellular fluid.

Values according to Katzung Table 27-1; Lexidrug monographs

Succinylcholine (SCh) Mechanism and Effects of Depolarizing NM Block

Initial stimulation with muscle contractions	**Phase I Block**	**Phase II Block**	**Later in Phase II Block**
SCh binds nAChRs ↓ ion channels open ↓ membranes are depolarized ↓ small spontaneous twitches (fasciculations) as Ca^{2+} is released from the sarcoplasmic reticulum	Depolarized membranes remain depolarized and unresponsive to subsequent impulses. Ca^{2+} is taken up by the sarcoplasmic reticulum ↓ flaccid paralysis = skeletal muscle relaxation	• Initial end plate depolarization decreases. • Membrane becomes repolarized. *However*, • the membrane can't easily be depolarized again because it is ***desensitized***. The channels behave as if they are in a prolonged closed state. (Mechanism is unclear.)	A non-sustained twitch response to a tetanic stimulus. The characteristics of the blockade are nearly identical to those of a nondepolarizing block.
Depolarizing block is augmented, not reversed, by cholinesterase inhibitors.		Prolonged exposure: large single dose, repeated dosing, continuous infusion	

Succinylcholine: Adverse Effects and Their Mechanisms

↑ Intragastric pressure	· Muscle fasciculations may increase risk of regurgitation and aspiration
↑ Intraocular pressure	· Muscle fasciculations may induce contraction of extra-ocular muscles with distortion and compression of the globe
↑ Intracranial pressure	· Transient. May result from stimulation of afferent muscle activity leading to increased cerebral flow.
Plasma cholinesterase deficiency Prolonged apnea	· Congenital: Homozygous for atypical *BCHE* gene · Acquired: Burn patients, anemia, decompensated heart disease, infections, malignant tumors, severe hepatic dysfunction, pregnancy, others
Hypersensitivity reaction	· Avoid in patients with a history of allergic reaction to succinylcholine or any component of the formulation
CONTRAINDICATIONS	· h/o malignant hyperthermia; myopathy / muscle paralysis; rhabdomyolysis; multiple trauma; spinal cord or upper motor neuron injury; major burns; ocular injury → **exaggerated hyperkalemic response**

Check your knowledge

- **What is the mechanism of muscle fasciculations caused by succinylcholine?**
- **List the serious adverse effects related to muscle fasciculations.**
- **What is a potential mechanism of the transient increased intracranial pressure caused by succinylcholine?**
- **What is the name of the byproduct of NMBA metabolism that in high concentrations is associated with lowering the seizure threshold?**

NMBAs

- Neuromuscular blocking agents (NMBAs) are highly polar compounds that cause temporary paralysis by blocking the transmission of nerve impulses to the muscles by actin on nicotinic receptors in the neuromuscular junction.
- NMBAs are used to induce muscle relaxation to facilitate intubation, surgical procedures in conjunction with anesthesia, and in mechanical ventilation.
- Nondepolarizing agents, atracurium, cisatracurium, rocuronium, and vecuronium, are competitive inhibitors of ACh. They have durations of action from minutes to longer than an hour. Prolonged paralysis that can lead to respiratory paralysis (apnea). Bradycardia, hypotension, and anaphylaxis are other potential serious effects.
- Succinylcholine (SCh) is a depolarizing NMBA with a rapid onset and short duration of action. It binds the ACh receptors, initially causing channel opening, membrane depolarization and spontaneous muscle twitching. It maintains the receptors in the depolarized state, which are unresponsive to subsequent activation – Phase I Block. With prolonged exposure, the receptors become repolarized but desensitized to activation by ACh – Phase II Block.
- Succinylcholine may cause malignant hyperthermia, myopathy, rhabdomyolysis, hyperkalemia and cardiac dysrhythmia, bradycardia, increased intragastric, intraocular, and intracranial pressure,
- Cholinesterase inhibitors reverse block by the isoquinolines and aminosteroids. Sugammadex reverses rocuronium and vecuronium block by complexing the drugs in plasma. Depolarizing block by SCh reverses spontaneously as the drug diffuses out of the synapse.

Check your knowledge.

1. **What is the overall effect of ganglion blockade by mecamylamine?**

- Ganglion blockers suppress all autonomic outflow nAChRs of both parasympathetic and sympathetic ganglia and adrenal medulla, which suppresses all autonomic outflow, and suppress autonomic homeostatic reflexes that normally moderate autonomic responses: normal cardiac, vascular, neurologic, muscle, and neurohumoral reflexes

2. **What are the mechanism of orthostatic hypotension and symptoms caused by mecamylamine?**

- Orthostatic hypotension (a significant drop in blood pressure on standing) and syncope is pronounced because postural reflexes (sympathetic responses) that normally prevent venous pooling are abolished. Symptoms are dizziness, lightheadedness, and fainting (syncope).

3. **Why should mecamylamine be prescribed only for patients who are reliable and cooperative with the therapy?**

- Mecamylamine dosing can be complicated, therapy must be carefully monitored, and the drugs cause orthostatic hypotension. Patients should be cautioned about all potential side effects and drug interactions.

4. **Since mecamylamine is rarely used, why is it important for students to learn about it (other than because it might appear on a board exam?**

- Understanding the effects of ganglion blockers requires a deep knowledge of the autonomic nervous system (ANS). Grasping the concepts means students can appreciate how ganglion blockers influence various physiological processes and manage the potential side effects and clinical implications effectively.